

ML, PL, NLP
Boot camp

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Set - 20

100-5 Measure of
dispersion

100-6 sample variance
divided by $n-1$

100-7 standard
deviation

$\Rightarrow$ Methods of dispersion

• The main methods of
dispersion include

- ① Variance
- ② Standard deviation

~~Age~~ -

It is the one sample
distribution

Age = { 2, 2, 4, 4 }

$\rightarrow$ sample data

$$\text{Ages 2} = \{1, 1, 5, 5\}$$

$$\begin{aligned} \mu(\text{Ages 1}) &= \frac{2+2+4+4}{4} \\ &= \cancel{2} \frac{2}{4} = 3 \end{aligned}$$

$$\begin{aligned} \mu(\text{Ages 2}) &= \frac{1+1+5+5}{4} \\ &= \frac{2}{4} 3 = 3 \end{aligned}$$

⊕ here as we can see that even if data is different, the overall

→ But if we look carefully we can still see the ~~same~~ difference in the distance

→ this can be explained by variance

①

Variance

→ population data $\Rightarrow \{N\}$

sample data $= \{n\}$

population var

σ^2

$$= \sum_{i=1}^N \frac{(x_i - \mu)^2}{N}$$

↓

variance
deviation

↓
This is basically
the difference
of the square
of the ~~sum~~
~~of~~ ~~the~~ ~~data~~
number & the
mean

population mean $= \mu$

sample mean $= \bar{x}$

Let us take an example:-

$$\text{August 1} = \{2, 2, 4, 4\}$$

~~August 2 =~~

$$\mu = \frac{2+2+4+4}{4} = 3$$

x_i	μ	$(x_i - \mu)^2$	
2	3	$(2-3)^2$	1
2	3	$(2-3)^2$	1
4	3	$(4-3)^2$	1
4	3	$(4-3)^2$	1
			4

$$N = 4$$

$$\frac{4}{4} = 1$$

$$\therefore \sigma^2 = 1 \quad (\text{variance} = 1)$$

(5)

$\Rightarrow$ Now doing the same work for ages 2 column it can be done as:-

~~$\mu = \frac{1+1+5+5}{4} = 3$~~

$$\text{Ages 2} = \{1, 1, 5, 5\}$$

$$\mu = \frac{1+1+5+5}{4} = 3$$

x_i	μ	$(x_i - \mu)^2$
1	3	$(1-3)^2 = 4$
1	3	$(1-3)^2 = 4$
5	3	$(5-3)^2 = 4$
5	3	$(5-3)^2 = 4$
$N=4$		$\frac{16}{4} = 4$

$$\sigma^2 = 4 \text{ (variance = 4)}$$

$\Rightarrow$ Now talking about the sample variance, the formulae is:-

In Sample Variance
Formulae comes will
be

$$s^2 = \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{n-1} \right)$$

Now here in sample
variance, it's divided by
 $n-1$.

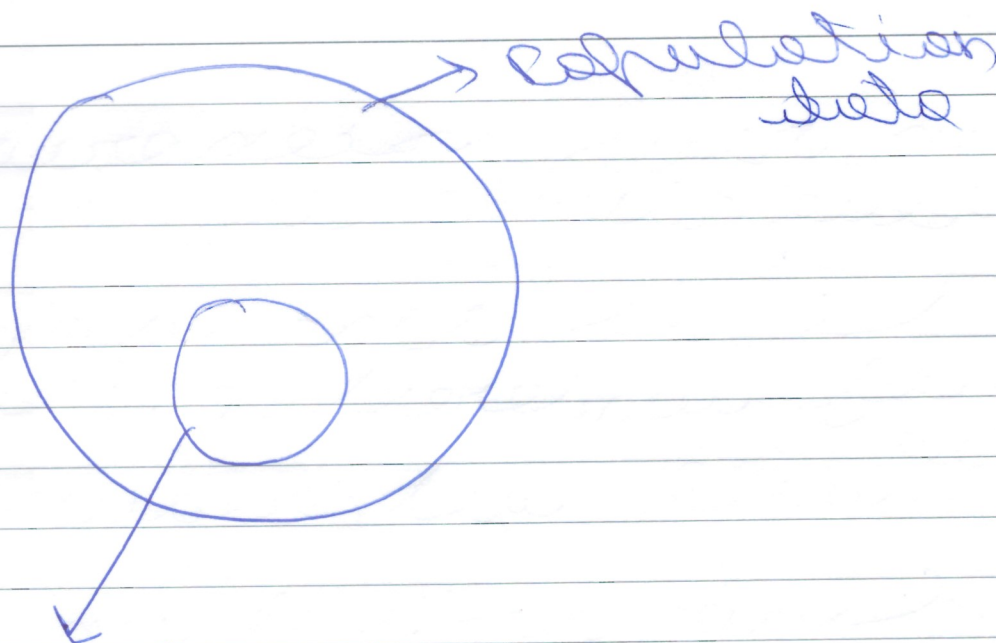
sample variance

$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$

In population variance
we divide by N

→ But in sample,
we divide by $(n-1)$

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sample data

We select the sample data & make inference about Population.

Suppose a variable of Age

Age = { --- }

Suppose we use the formula by

$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n}$$

If we use this formula we're underestimating the best population variance.

In this scenario
we divide by $n-1$,

→ as the gap
of underestimation
will get reduced.

→ We're not under
estimating true populat
ion variance

$$s^2 = \sum_{i=1}^n \frac{(x - \bar{x})^2}{n-1}$$

→ as shown as

Bessel's correct

Degree of freedom = $n-1$

Standard Deviation

Population
Data
N

$$\mu = \sum_{i=1}^N \frac{(x_i)}{N}$$

$$\sigma^2 = \sum_{i=1}^N \frac{(x_i - \mu)^2}{N}$$

$$\sqrt{\sigma^2}$$

Population
Standard Deviation

It tells that how far up or down data points is away from mean

Sample
Data
n

$$\bar{x} = \sum_{i=1}^n \frac{(x_i)}{n}$$

$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$

$$\sqrt{s^2}$$

~~Sample~~ Sample
Standard
Deviation

$$s = \sqrt{\text{Sample variance}}$$